Stoichiometry: Moles, Mass and Balanced Equations

My General Chemistry 1 reviewer on stoichiometry. Stoichiometry is the part of chemistry that deals with the number relationships between reactants and products in a chemical reaction. Think of it as following a recipe for pandesal, but with atoms.

The mole concept

Atoms are far too small to count one by one, so chemists count them in groups called moles. One mole is the amount of substance that contains exactly 6.022 x 10^23 particles. This number is called Avogadro's number, named after the Italian scientist Amedeo Avogadro.

The particles can be atoms, molecules, ions or formula units, depending on the substance.

- 1 mol of carbon atoms = 6.022 x 10^23 carbon atoms
- 1 mol of water molecules = 6.022 x 10^23 H2O molecules
- To convert moles to particles, multiply by 6.022 x 10^23. To go from particles to moles, divide.

Just as a dozen always means 12, a mole always means 6.022 x 10^23.

Molar mass

The molar mass of a substance is the mass in grams of one mole of it, with units of g/mol. For an element, it is numerically equal to the average atomic mass on the periodic table. For a compound, add up the atomic masses of all the atoms in the formula.

- H = 1.01 g/mol, C = 12.01 g/mol, O = 16.00 g/mol, Na = 22.99 g/mol, Cl = 35.45 g/mol
- H2O = 2(1.01) + 16.00 = 18.02 g/mol
- NaCl (table salt, asin) = 22.99 + 35.45 = 58.44 g/mol
- CO2 = 12.01 + 2(16.00) = 44.01 g/mol

Key conversions:

- moles = mass / molar mass
- mass = moles x molar mass

Balanced equations and mole ratios

A chemical equation must be balanced because of the law of conservation of mass: atoms are not created or destroyed in a chemical reaction, only rearranged. Balancing means adjusting the coefficients (the big numbers in front of formulas), never the subscripts.

Example: 2H2 + O2 -> 2H2O

The coefficients tell us the mole ratio. Here, 2 mol of H2 react with 1 mol of O2 to produce 2 mol of H2O. The mole ratio is the bridge in every stoichiometry problem. The usual path is:

- grams of A -> moles of A (divide by molar mass)
- moles of A -> moles of B (use the mole ratio from the balanced equation)
- moles of B -> grams of B (multiply by molar mass)

Important: coefficients relate moles, not grams. You cannot use the ratio directly on masses.

Limiting and excess reactants

In real reactions, reactants are rarely mixed in the perfect ratio. The limiting reactant is the one that is used up first, and it decides the maximum amount of product that can form. The excess reactant is the one left over after the reaction stops.

Kitchen analogy: if one sandwich needs 2 slices of bread and 1 slice of cheese, and you have 10 slices of bread but only 3 slices of cheese, you can make only 3 sandwiches. Cheese is limiting; bread is in excess, with 4 slices left.

To find the limiting reactant, convert each reactant to moles, then check how much of the other reactant each one would need (or compute how much product each could make). The one that gives less product is limiting.

Theoretical, actual and percent yield

- Theoretical yield is the maximum amount of product calculated from the limiting reactant, assuming a perfect reaction.
- Actual yield is the amount you really obtain in the laboratory. It is usually less because of side reactions, incomplete reactions, or product lost during transfer and filtering.
- Percent yield = (actual yield / theoretical yield) x 100%

Fully worked example

Problem: 8.08 g of H2 reacts with 48.00 g of O2 to form water. Find the limiting reactant, the theoretical yield of water, the excess left, and the percent yield if 48.65 g of water was actually collected.

Equation: 2H2 + O2 -> 2H2O

Step 1. Convert to moles.

- H2: 8.08 g / 2.02 g/mol = 4.00 mol
- O2: 48.00 g / 32.00 g/mol = 1.50 mol

Step 2. Find the limiting reactant. 4.00 mol H2 would need 4.00 / 2 = 2.00 mol O2. We only have 1.50 mol O2, so O2 is the limiting reactant and H2 is in excess.

Step 3. Theoretical yield from O2. The ratio is 1 mol O2 : 2 mol H2O.

- 1.50 mol O2 x 2 = 3.00 mol H2O
- 3.00 mol x 18.02 g/mol = 54.06 g H2O

Step 4. Excess H2. H2 used = 1.50 mol O2 x 2 = 3.00 mol. H2 left = 4.00 - 3.00 = 1.00 mol, which is 1.00 x 2.02 = 2.02 g.

Step 5. Percent yield = (48.65 / 54.06) x 100% = 90.0%.

Quick check

- Q: What is the value of Avogadro's number? A: 6.022 x 10^23 particles per mole.
- Q: What is the molar mass of CO2? A: 44.01 g/mol.
- Q: Which reactant determines the theoretical yield? A: The limiting reactant.
- Q: When balancing an equation, what may you change? A: Only the coefficients, never the subscripts.

- Q: If the theoretical yield is 20.0 g and the actual yield is 15.0 g, what is the percent yield? A: 75.0%.